Technical University of Munich
Max Planck Institute of Quantum Optics

An atom-photon quantum gate using an atomic clock qubit and a birrefringent cavity

Leart Zuka

Thesis submitted within the M.Sc. Condensed Matter Physics

Supervisor: Prof. Dr. Gerhard Rempe
Second Examiner:

March 10, 2026

Abstract

Single atoms coupled to optical cavities provide a powerful platform for photonic quantum information processing. Minimizing the influence of external factors like magnetic field fluctuations is a key goal in this endeavor. In this work, I investigate a novel gate protocol that employs a single ^{87}Rb atom prepared in a magnetic-field-insensitive clock qubit and coupled to a high-finesse birefringent FabryPerot cavity. The cavitys high birefringence creates two well-separated polarization eigenmodes, enabling polarization-selective reflectivity that depends on the atomic qubit state. This allows for the implementation of a controlled phase (CPHASE) gate between the atomic clock states and our photonic qubits. By rotating the basis of the photonic qubit, one is then able to realize an atom-photon controlled-NOT (CNOT) gate.

Show some results here and talk about how well it works and also how one might be able to use the gate

I present simulations based on inputoutput theory and master-equation modeling that quantify the conditional reflection amplitudes, gate truth table, and resulting fidelities under realistic experimental imperfections such as finite mode matching, cavity loss channels, and multiphoton contributions. Our results indicate that atom-state-dependent phase shifts on the photonic qubit are achievable in the current system, providing a viable path toward a robust, high-fidelity, cavity-assisted atomphoton quantum gate.

Contents

Chapter 1

Introduction

Bla bla bla quantum gate important for distant computation between two nodes very cool. Cite some stuff from kimble and duan here [1], talk about recent advancements or stuff that was similar(cite reiserer here [2]).
Basically the idea here is to have a good introduction into the topic. Here I can then talk about:
- what makes up quantum computation - why it's so difficult to scale up in a certain way (maybe look at different examples with large amounts of ^{87}Rb atoms and comment on what the challenges there are when it comes to first of all having lots of them, and them keeping them coherent and stuff like that). - Talk basically a lot about what Duan and Kimble talked about in their introduction of the paper since they were talking a lot about how computing at one node is hard, but spliting it up into smaller nodes and thus making it able to compute stuff. - Then talk about how one is then able to connect the nodes with the gate approach. - Explain what makes it unique (aka the birefringence of the cavity and the ability to then use the clock states which makes it insensitive to external magnetic fields) - Give an outlook over the chapters that are to follow

Chapter 2

Theory

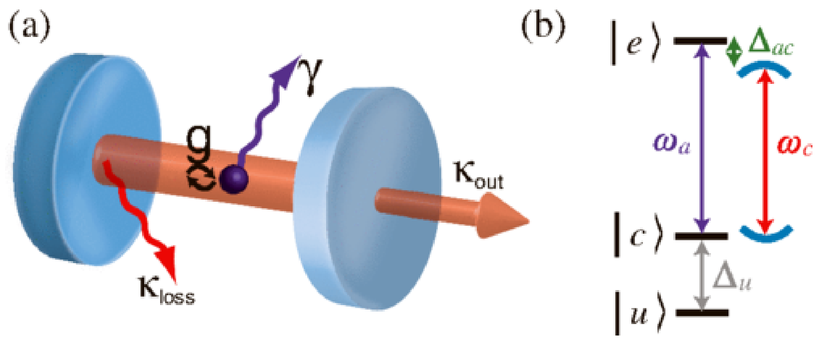
What follows is going to be the introduction and explanation of specific concepts required to understand the key points of this thesis.

2.1 Cavity Quantum Electrodynamics

The field of *Cavity Quantum Electrodynamics* explains the interaction between a resonator (in this case a Fabry-Perot cavity), and a piece of matter with which the light inside the resonator interacts with (here: a ^{87}Rb atom). By placing the atom between two highly-reflective mirrors forming the optical cavity, one increases the amount of times a photon that is inside a cavity bounces back and forth between the two mirrors. Increasing the amount of bounces thus effectively increases the cross-section by the number of round-trips the photon takes, allowing for a strong light-matter interaction.[3]
The theoretical model which explains this interaction in a fully quantum mechanical picture is called the *Jaynes-Cummings Model* [4] and was named after Edwin Jaynes and Fred Cummings who developed it in 1963. For a first look into this model, we'll be considering a three-level-emitter (i.e. our atom), which is interacting with a single cavity optical mode.
The *Jaynes-Cummings* (JC) Model, takes many different quantities into consideration when explaining the interaction and its strength between light and matter.
The photon located in the cavity (which supports only a single optical mode), has a frequency of ω_c. Our atom, which is also located in the cavity, supports a transition frequency close to ω_c, from a coupled state $|c\rangle$ and an excited state $|e\rangle$ at a frequency ω_a, which is described in the dipole approximation. In order to then complete our three-level-emitter picture, we include an uncoupled state $|u\rangle$ with which the photon does not interact due to this state not being coupled to the cavity light mode.
Lastly, any real atom-cavity system, additionally suffers from losses, those mainly being the spontaneous decay of the atomic excitation at a rate of 2γ, and the damping of the light inside our cavity through the absorption or out-coupling by one of the cavity mirrors at a combined rate of κ.
The full Hamiltonian which describes interaction between the atom and the cavity thus reads [5]

$$\mathcal{H}_{\mathcal{JC}} = \underbrace{\hbar\omega_c\hat{a}^\dagger\hat{a}}_{\widehat{=}\text{cavity field}} + \underbrace{\hbar\omega_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce}}_{\widehat{=}\text{atom}} + \underbrace{\hbar g(\hat{a}\hat{\sigma}_{ce}^\dagger + \hat{a}^\dagger\hat{\sigma}_{ce})}_{\widehat{=}\text{light-atom interaction}} \tag{2.1}$$

The three terms in (2.1) describe the energy of the cavity field, the energy of a bare atom, and the coherent coupling between the two governed by a coupling strength g.
The operators $a^\dagger$ and a are the creation and annihilation operators of the cavity light modes , whereas $\hat{\sigma}_{ce} = |c\rangle\langle e|$ and $\hat{\sigma}_{ce}^\dagger = |e\rangle\langle c|$ are the atomic raising and lowering operators, respectively.
The form of (2.1) assumes the so called *Rotating-Wave Approximation* in which fast oscillating terms at optical frequencies can be neglected.
Solving for the eigenenergies of the system one is left with:



$$E_{\pm}(n) = n\hbar\omega_c + \frac{1}{2}\hbar\omega_a \pm \hbar\sqrt{ng^2 + \frac{1}{4}\Delta_{ac}^2}, \quad n \in \mathbb{N}_0 \tag{2.2}$$

The main take-away from (2.2) is that on resonance ($\omega_a = \omega_c$), the eigenenergies of the system are split by $2\hbar g\sqrt{n}$, giving rise to the so-called *Vacuum Rabi Splitting*.

2.1.1 Birefringent Cavity

Equation (2.1) only covers the case for having a single cavity light-mode. The present example however features a birefringent cavity, which is achieved by physically changing the geometric shape of the cavity mirror during production of the fiber ends.[6] This forces a change in the optical path lengths between different polarizations, leading to a frequency splitting $\Delta\nu$ proportional to the ellipticity ϵ of the mirror surfaces:[7]

$$\Delta\nu = \frac{\nu_{FSR}}{2\pi k}\frac{\epsilon^2}{R_2}. \tag{2.3}$$

The property ν_{FSR} in (2.3) describes the free spectral range, and $k = \frac{2\pi}{\lambda}$ the photon wavenumber. This means that we need to consider a system with 2 different light modes for our JC Hamiltonian, which in our case are going to be the π and the V modes, whereas we're going to be coupled to the π mode, and the V mode will be detuned by ~ 500 MHz. This extra consideration can be included in the Hamiltonian as such:

$$H_{JC} = \hbar\omega_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar(\omega_{c,\pi} + 0.5)\hat{a}_V^\dagger\hat{a}_V + \hbar\omega_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} + \hbar g(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}) \tag{2.4}$$

2.1.2 External Drive

At the heart of the atom-photon gate lies the reflection of the incoming photon off of the cavity system. In order to describe this in a quantum mechanical way, one needs to include a hamiltonian, namely a driving hamiltonian $H_{D,S} = i(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(\hat{a} + \hat{a}^\dagger)$. This hamiltonian $H_{D,S}$ is defined in the Schrödinger picture, and has a complex drive amplitude ϵ and driving field frequency ω_d. Adding this extra term to (2.4), and changing into the rotating frame of the driving field, one gets the following Hamiltonian:

$$\begin{aligned} H_{JC} =& \hbar\Delta_{c,\pi}\hat{a}_\pi^\dagger\hat{a}_\pi + \hbar\left(\Delta_{c,\pi} + 0.5\right)\hat{a}_V^\dagger\hat{a}_V + \hbar\Delta_a\hat{\sigma}_{ce}^\dagger\hat{\sigma}_{ce} \\ &+ \hbar g\left(\hat{a}_\pi\hat{\sigma}_{ce}^\dagger + \hat{a}_\pi^\dagger\hat{\sigma}_{ce} + \hat{a}_V\hat{\sigma}_{ce}^\dagger + \hat{a}_V^\dagger\hat{\sigma}_{ce}\right) \\ &+ \mathrm{i}\left(\hat{a}_{\pi,V}\epsilon^* + \hat{a}_{\pi,V}^\dagger\epsilon\right) \end{aligned} \tag{2.5}$$

This then gives a definition of the JC Hamiltonian with respect to an external driving field, and based on the detunings of the driving field with respect to the cavity resonance frequency $\Delta_{c,\pi} = \omega_{c,\pi} - \omega_d$ and the atomic transition frequency $\Delta_a = \omega_a - \omega_d$, respectively.

2.2 Reflection from a Cavity-Atom System

For us it's important that the information we send in via the photon, won't get lost and will get reflected, thus ensuring that our atom-photon gate is suitable for quantum information processing. Therefor when the light enters our cavity, one must ensure by using proper system parameters that the light has maximum transmission through the first cavity mirror (called outcoupling (OC) mirror) relative to the opposing highly reflective (HR) mirror. The highly reflective mirror on the other hand must have maximum reflection so that the information stored in the photon doesn't get lost after subsequent bounces inside the cavity while it's trying to exit via the outcoupling mirror. The total losses can thus be divided into the following parts:

$$\kappa = \kappa_{oc} + \kappa_{HR} + \kappa_{paras} \tag{2.6}$$

The escape probability of our photon is defined by the outcoupling ratio $\frac{\kappa_{oc}}{\kappa}$. This quantity was measured and is $\sim 80\%$.
Another quantity is the optical mode matching between the reflected light at the first cavity and the incoming light (μ_{fr}) aswell as the optical modes of the incoming light and the optical mode inside of the cavity (μ_{fc}).

2.2.1 Input-Output Theory

Our photon that is going to be reflected off of our cavity system can be viewed as an input field. This then can be linked to an output field via input-output formalism. When driving the system only from one side, as it is the case here, the relation between input and output becomes:[8]

$$a_{out}(t) = \mu_{fr} a_{in}(t) + \mu_{fc}\sqrt{\kappa_{oc}}\langle a\rangle(t) \tag{2.7}$$

Here (2.7) was adjusted to also consider the mode matching parameters. Solving Langevin equations for neglidgable atomic excitation and photonic wave envelopes which vary only on a time scale longer than κ, one can write down the analytical representation of the reflection amplitude[9, 10, 11]:

$$r(\omega) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}(i\Delta_a + \gamma)}{(i\Delta_c + \kappa)(i\Delta_a + \gamma) + g^2} \tag{2.8}$$

The reflection amplitude in (2.8) depends on the detuning $\Delta_{c,a} = \omega - \omega_{c,a}$ between the driving laser frequency and the cavity resonance frequency ω_c or atomic transition frequency ω_a.
Include reflection spectrum for different detunings and also phase across different detunings

2.3 Conditional Reflection

Looking at (2.8) one can see that the behavior of the reflection amplitude at zero detuning ($\Delta_a = \Delta_c = 0$) is:

$$r(\Delta = 0) = \mu_{fr} - \mu_{fc}^2 \frac{2\kappa_{oc}\gamma}{\kappa\gamma + g^2} \tag{2.9}$$

In the event of an uncoupled atom-cavity system ($g = 0$), which would be achieved by preparing the atom in the $|u\rangle$ state, the reflection coefficient r is going to be smaller than 0, which would imply a relative phase shift of π between the reflected and incoming light field. Placing the atom in the coupled state $|c\rangle$ however, the reflection coefficient r is going to be bigger than 0 (at sufficiently high coopoerativity). Equipped with this knowledge, that there is a relative phase shift of the output field depending on in which state the atom is, allows the realisation of quantum atom-photon gate with the atom being the controlling qubit and the photon being the target qubit.[1]

2.3.1 Reflection in CPF basis

Changing the nomenclature to match that regularly used in quantum computing, the uncoupled state of the atom is going to be the $|0\rangle$ state, and the coupled one the $|1\rangle$ state. Our photon which as stated before, is going to be either π-polarized or V-polarized, will thus be represented as the input states $|\pi\rangle$ and $|V\rangle$. For the case of our atom being in the $|0\rangle$ state and us sending in a photon which is resonant to the π-mode of the cavity (here: $|\pi\rangle$), the output-field will receive a phase flip, an none for all the other cases either due to being far detuned from the cavity resonance (case of sending $|V\rangle$) and/or atom being in the coupled case $|1\rangle$ and the photon thus getting reflected off of the first cavity mirror:

$$
\begin{aligned}
|0,\pi\rangle &\to -|0,\pi\rangle \\
|0,V\rangle &\to |0,V\rangle \\
|1,\pi\rangle &\to |1,\pi\rangle \\
|1,V\rangle &\to |1,V\rangle
\end{aligned}
$$

From this we deduce that output from the atom-photon reflection gives us a CPF gate. This conditional phase shift allows the construction of a universal quantum gate that can be transformed into any two-qubit gate using rotations of the individual qubits, which in the case of the photon can be done via retardation waveplates.

2.3.2 Reflection in CNOT basis

Rotating the photonic basis states to

$$
\begin{aligned}
|+\rangle &= \frac{|V\rangle + |\pi\rangle}{\sqrt{2}} \\
|-\rangle &= \frac{|V\rangle - |\pi\rangle}{\sqrt{2}}
\end{aligned}
$$

and keeping the atom either in the coupled ($|1\rangle$) or uncoupled case ($|0\rangle$), results in the following reflection:

$$
\begin{aligned}
|0,+\rangle &\to |0,-\rangle \\
|0,-\rangle &\to |0,+\rangle \\
|1,+\rangle &\to |1,-\rangle \\
|1,+\rangle &\to |1,-\rangle
\end{aligned}
$$

This then tells us, that simply by using a superposition of our CPF basis, our CPF gate now represents an atom-photon controlled-NOT (CNOT) gate.
Maybe also include brewster plate stuff here, although not quite sure

Chapter 3

Experimental Setup

This chapter will go through the experimental setup that was used to construct the atom-photon gate, and the reasoning behind some components that were chosen due to achieve the highest fidelity.

3.1 Overview of the beam path

3.2 Birefringent Cavity

3.2.1 Cavity Geometry and Birefringence

3.2.2 Cavity Spectroscopy

3.3 Polarization Management and Loss Compensation

3.3.1 Brewster Plate Array

3.3.1.1 Custom Brewster Plate Mount

3.3.1.2 Calibration of Brewster Plates

3.3.2 Pellicle Beamsplitter

Appendix A

Analytical Derivations

A.1 Output Light Analysis

A.1.1 Problem

From our QuTip simulation we're able to calculate the output field $a_{out}(t)$ via input output theory:

$$a_{out}(t) = \mu_{fr} a_{in}(t) + \mu_{fc}\sqrt{\kappa_{oc}}\langle a\rangle(t) \tag{A.1}$$

Where all the fields mentioned in (A.1), can be in one of the following bases: $\{|\pi\rangle, |V\rangle, |+\rangle, |-\rangle\}$. Now the question becomes, how is one able to analyze this output field and generate a truth table from this. For that we're going to calculate the mean photon number in the output field via:

$$\bar{n}_{out;\{|\pi\rangle,|V\rangle,|+\rangle,|-\rangle\}} = \int |a_{out;\{|\pi\rangle,|V\rangle,|+\rangle,|-\rangle\}}|^2 dt$$

For the problem we're going to be looking at, we're only going to consider the $|+\rangle$ and the $|-\rangle$ modes of our light, as these are the polarizations which are set in our CNOT basis, for which we want to calculate the fidelity for.

A.1.2 Analysis via detector

Now after having extracted the mean photon number in our output field for our two polarizations we need to consider the framework which we're using to analyze this output field. In our case that is the case of 2 independent detectors, where one detects the $|+\rangle$ polarized light and the other detects the $|-\rangle$ polarized light. Both detectors share the same efficiency η, and also the same dark count rate p_{dc}, which is the rate at with a single detector will fire even upon receiving no signal. The efficiency can be viewed as a beam splitter. Meaning, that if we were to have $\bar{n}$ photon in from of our detector that has an efficiency η, this can be treated as a perfect detector with $\eta = 1$ but with a beam splitter in front of our perfect detector that will "take" away $(1-\eta)\bar{n}$ photons from our initial pulse. The rest of the icoming pulse (now: $\eta\bar{n}$) will however get detected perfectly. Since we're only interested in the cases where we have 1 detector event ($k = 1$) upon receiving $\lambda = \eta\bar{n}$ photons, we can treat this problem via poisson statistics with:

$$P(k, \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} = \lambda e^{-\lambda}$$

This however doesn't take into consideration our dark counts yet.

A.1.3 Detector statistics

If our detector fires, and gives us 1 detector click, it can come from 2 sources:

- a dark count
- a photon

This we can then group up into 4 cases for each of the 2 possible sources of a click

- $P(0 \text{ Photons, } 0 \text{ dark counts}) = e^{-\lambda} \cdot e^{-p_{dc}} = e^{-(\lambda + p_{dc})}$
- $P(1 \text{ Photon, } 0 \text{ dark counts}) = \lambda e^{-\lambda} \cdot e^{-p_{dc}} = \lambda e^{-(\lambda + p_{dc})}$
- $P(0 \text{ Photons, } 1 \text{ dark count}) = e^{-\lambda} \cdot p_{dc} e^{-p_{dc}} = p_{dc} e^{-(\lambda + p_{pc})}$
- $P(1 \text{ Photons, } 1 \text{ dark counts}) =$ Can be ignored as this would produce 2 clicks, which we're post selecting away

Meaning that we have 2 out of our 4 possible events which then produce only 1 click, giving us the probability that each of our detectors produces 1 click:

$$P_{\pm,click} = (\lambda_{\pm} + p_{dc}) e^{-(\lambda_{\pm} + p_{dc})} \tag{A.2}$$

A.1.4 2 Detector Statistics

As mentioned before, we have 2 detectors with the same efficiency η and the same detector dark rate p_{dc}. These can produce a click independent from one another, but we're only interested in the cases where **ONLY** 1 produces a click, as the photon can (ideally) only be either $|+\rangle$ or $|-\rangle$ polarized.

$$\begin{aligned} P(\text{only}|+\rangle) &= P_{+,click} \cdot P_{-,0clicks} \\ &= (\lambda_+ + p_{dc}) e^{-(\eta(\bar{n}_{out,+} + \bar{n}_{out,-}) + 2p_{dc})} \end{aligned}$$

$$P(\text{only}|-\rangle) = (\lambda_- + p_{dc}) e^{-(\eta(\bar{n}_{out,+} + \bar{n}_{out,-}) + 2p_{dc})}$$

Adding the two together gives us the probability that at just from the fact that only one detector clicked, something sort of useful came out of our output field, which we're going to call P_{valid}:

$$P_{valid} = (\lambda_+ + \lambda_- + 2p_{dc}) e^{-(\eta(\bar{n}_{out,+} + \bar{n}_{out,-}) + 2p_{dc})}$$

However now we need to filter out only the cases where the detector event was actually triggered by a photon and not a dark count.

A.1.5 Photon Event Filtering

Here the probability that the 1 detector event in **ONLY** was caused by a photon is:

$$\begin{aligned} P(1 \text{ photon in } + \text{ \& no click in -}) &= \lambda_+ e^{-(\lambda_+ + p_{dc})} \cdot e^{-(\lambda_- + p_{dc})} \\ &= \lambda_+ e^{-(\eta(\bar{n}_{out,+} + \bar{n}_{out,-}) + 2p_{dc})} \\ P(1 \text{ photon in - \& no click in } +) &= \lambda_- e^{-(\eta(\bar{n}_{out,+} + \bar{n}_{out,-}) + 2p_{dc})} \end{aligned}$$

Thus giving us the probability that 1 click was produced in only 1 of the two detectors:

$$P_{only} = (\lambda_+ + \lambda_-) e^{-(\eta(\bar{n}_{out,+} + \bar{n}_{out,-}) + 2p_{dc})}$$

This then allows us now to calculate how many of our valid clicks, were actually produced by only 1 photon event going only to 1 of the 2 detectors, essentially giving us a sort of signal-to-noise ratio:

$$P_{SNR} = \frac{P_{only}}{P_{valid}} = \frac{\lambda_+ + \lambda_-}{\lambda_+ + \lambda_- + 2p_{dc}}$$

A.1.6 Photon Distribution

All that's left now is to calculate how much of our light went to the "+"-detector, and how much went to our "-"-detector, which is just like calculating the ratios of all of our events that were triggered only by photons, and the probability of the respective detector itself:

$$R_+ = \frac{P(\text{1 photon in + and no click in -})}{P_{only}} = \frac{\bar{n}_{out,+}}{\bar{n}_{out,+} + \bar{n}_{out,-}}$$

$$R_- = \frac{P(\text{1 photon in - and no click in +})}{P_{only}} = \frac{\bar{n}_{out,-}}{\bar{n}_{out,+} + \bar{n}_{out,-}}$$

What then ends up going into our truth table is the following:

- For the $|+\rangle$-part $= R_+ \cdot P_{SNR} + (1 - P_{SNR}) \cdot 0.5$
- For the $|-\rangle$-part $= R_- \cdot P_{SNR} + (1 - P_{SNR}) \cdot 0.5$

A.2 Driving Hamiltonian

Since we're mainly going to be working with the *mesolve* function in QuTip to simulate the dynamics of the atom-photon gate it's imperative to know what each and every part does. Next to the Jaynes-Cummings-Hamiltonian the most important part is the one explaining the *photon that gets reflected from the cavity*, most notably the part before it gets reflected aka the *driving Hamiltonian* H_d. For this we're going to just write down once, the dynamics of a coherent driving field which will be the foundation of my photon that shall or shall not enter the cavity.

The photon is going to come from a laser, that is going to be described as a classical oscillating electric field. That light field has an amplitude E_0 and a frequency ω_d:

$$E(t) = E_0 \cos(\omega_d t)$$

Using the euler formula we can rewrite this to:

$$E(t) = \frac{E_0}{2}(e^{i\omega_d t} + e^{-i\omega_d t})$$

The light field operator $\hat{E}$ interacts with the light in our cavity (which get described by our creation and annihilation operators) linearly:

$$\hat{E} \propto (a + a^\dagger)$$

When now describing the *driving Hamiltonian* what we're doing is just overlaping the light field operator with the oscillating electric field:

$$H_{drive} \propto E(t) \cdot (a + a^\dagger)$$
$$H_{drive} = (e^{i\omega_d t} + e^{-i\omega_d t}) \cdot (a + a^\dagger)$$

From this we get four terms where two cancel out as they aren't energy conserving rendering us with the final expression for our driving Hamiltonian:

$$\boxed{H_{drive} = a^\dagger e^{-i\omega_d t} + a e^{+i\omega_d t}}$$

Where the first term adds a photon to our cavity and the other removes one. Sometimes you'll see a minus instead of a plus between the two terms, this is just due to different conventions of the light field, where people chose a sin to describe the oscillating part instead of a cos, but the physics remain the same.

A.3 Analytical derivation for case of atom in $|0\rangle$ and photon is in $|h\rangle$

Our Hamiltonian is as follows:

$$H = \hbar g(|e\rangle\langle 1| a_h + |1\rangle\langle e| a_h^\dagger)$$

We drive the a_h mode of our cavity through $a_h^{in}(t)$, thus giving us the following equation of motion for our a_h cavity mode:

$$\dot{a_h} = -i[a_h, H] - (i\Delta + \frac{\kappa}{2})a_h - \sqrt{\kappa}a_h^{in}(t)$$

Because a_h commutes with H, the commutator is 0 and we get

$$\rightarrow \dot{a_h} = -(i\Delta + \frac{\kappa}{2})a_h - \sqrt{\kappa}a_h^{in}(t)$$

We now drop the subscript h as we'll only be looking at that case.
Multiplying with $e^{i\omega t}$, and integrating we get:

$$\int dte^{i\omega t}\dot{a}(t) = \underbrace{[e^{i\omega t}a(t)]_{-\infty}^{\infty}}_{=0} - \underbrace{i\omega\int dte^{i\omega t}a(t)}_{=i\omega a(\omega)}$$

where in the last expression we used that the integral performs a Fourier transformation.

$$\begin{aligned}\Rightarrow -i\omega a(\omega) &= -(i\Delta + \frac{\kappa}{2})a(\omega) - \sqrt{\kappa}a^{in}(\omega)\\ \sqrt{\kappa}a^{in}(\omega) &= -(i\Delta + \frac{\kappa}{2} - i\omega)a(\omega)\\ \rightarrow a(\omega) &= -\frac{\sqrt{\kappa}}{i\Delta + \frac{\kappa}{2} - i\omega}a^{in}(\omega)\end{aligned}$$

Now from our regular input output relation

$$a^{out}(\omega) = a^{in}(\omega) + \kappa a(\omega)$$

We thus get:

$$\begin{aligned}a^{out}(\omega) &= a^{in}(\omega) - \frac{\sqrt{\kappa}}{i\Delta + \frac{\kappa}{2} - i\omega}a^{in}(\omega)\\ a^{out}(\omega) &= (1 - \frac{\kappa}{i\Delta + \frac{\kappa}{2} - i\omega})a^{in}(\omega)\\ a^{out}(\omega) &= (1 - \frac{i\Delta + \frac{\kappa}{2} - i\omega - \kappa}{i\Delta + \frac{\kappa}{2} - i\omega})a^{in}(\omega)\\ a^{out}(\omega) &= (\frac{i\Delta - \frac{\kappa}{2} - i\omega}{i\Delta + \frac{\kappa}{2} - i\omega})a^{in}(\omega)\\ &= \underbrace{\frac{i(\Delta - \omega) - \frac{\kappa}{2}}{i(\Delta - \omega) + \frac{\kappa}{2}}}_{h(\omega)}a^{in}(\omega)\end{aligned}$$

We can then use the convolution theorem in order to solve for $a^{out}(t)$

$$a^{out}(t) = \int_{-\infty}^{\infty} d\tau \underbrace{h(t-\tau)}_{h(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega e^{-i\omega t} h(\omega)} a^{in}(\tau)$$

Now using that $h(\omega) = 1 - \frac{\kappa}{i(\Delta-\omega)+\frac{\kappa}{2}}$ we get for $h(t)$ (via partial integration):

$$\begin{aligned} h(t) &= \delta(t) - \kappa e^{-i\delta t} e^{-\frac{\kappa}{2}t}\theta(t) \\ \Rightarrow a^{out}(t) &= a^{in}(t) - \kappa \int_{-\infty}^{\infty} d\tau e^{-i\delta(t-\tau)} e^{-\frac{\kappa}{2}(t-\tau)} a^{in}(t) \end{aligned}$$

For slowly varying pulses we can now approximate that $a^{in}(\tau) \approx a^{in}(t)$ thus giving us:

$$\begin{aligned} \Rightarrow a^{out}(t) &\approx a^{in}(t) - \frac{\kappa}{i\Delta + \frac{\kappa}{2}} a^{in}(t) \\ &= \frac{i\Delta - \frac{\kappa}{2}}{i\Delta + \frac{\kappa}{2}} a^{in}(t) \end{aligned}$$

Which for no detuning (so $\Delta = 0$) gives us:

$$a^{out}(t) = -a^{in}(t) \tag{A.3}$$

A.3.1 Lindblad Formalism

We start from the master equation:

$$\dot{\rho} = -\frac{i}{\hbar}[H,\rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \right), \tag{A.4}$$

Now, for an operator O,

$$\frac{d}{dt}\langle O\rangle = \mathrm{Tr}\,(O\dot{\rho}) \tag{A.5}$$

$$= \mathrm{Tr}\left(O \left[-\frac{i}{\hbar}[H,\rho] + \sum_j \gamma_j \left(L_j \rho L_j^\dagger - \tfrac{1}{2}\{L_j^\dagger L_j, \rho\} \right) \right] \right). \tag{A.6}$$

The Hamiltonian contribution gives

$$-\mathrm{Tr}\left(O \frac{i}{\hbar}[H,\rho] \right) = -\frac{i}{\hbar}\,\mathrm{Tr}\,(OH\rho - O\rho H) = -\frac{i}{\hbar}\,\mathrm{Tr}\,([O,H]\rho) = -\frac{i}{\hbar}\langle [O,H]\rangle. \tag{A.7}$$

For the dissipative terms we obtain

$$\sum_j \gamma_j\,\mathrm{Tr}\left(O L_j \rho L_j^\dagger - \tfrac{1}{2} O L_j^\dagger L_j \rho - \tfrac{1}{2} O \rho L_j^\dagger L_j \right). \tag{A.8}$$

Using cyclicity of the trace:

$$= \sum_j \gamma_j \left(\langle L_j^\dagger O L_j \rangle - \tfrac{1}{2}\langle \{L_j^\dagger L_j, O\}\rangle \right). \tag{A.9}$$

Therefore the general equation of motion is

$$\frac{d}{dt}\langle O\rangle = -\frac{i}{\hbar}\langle [O,H]\rangle + \sum_j \gamma_j \left(\langle L_j^\dagger O L_j \rangle - \tfrac{1}{2}\langle \{L_j^\dagger L_j, O\}\rangle \right). \tag{A.10}$$

A.3.2 Calculation of the Maxwell Bloch Equations (no lossy channels)

A.3.2.1 Jaynes Cummings Hamiltonian

Assuming ground state has zero energy our Jaynes-Cummings Hamiltonian is equal to:

$$H_{JC,S} = \underbrace{\omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma}_{=H_0,S} + \underbrace{g(\sigma + \sigma^\dagger)(a + a^\dagger)}_{=H_{int,S}} + H_{drive,S} \tag{A.11}$$

where σ is the atomic lowering operator,ω_a the atomic transition frequency between the levels $|1\rangle$ and $|e\rangle$,a the photonic destruction operator, and ω_a the cavity mode resonance frequency.
H_{drive} in this case is the external light field which is driving our system, which in the semiclassical formalism has the following form [12]:

$$H_{drive,S} = i(\epsilon e^{-i\omega_d t} + \epsilon^* e^{i\omega_d t})(a + a^\dagger) \tag{A.12}$$

Here, a again is the photonic destruction operator, ω_d is the frequency of the driving field, and ϵ is the complex amplitude of our incoming pulse.
In general going from the Schrödinger picture to the interaction picture, is achieved by converting our operators into the interaction picture via following equation:

$$i\frac{dA_I}{dt} = [A_I(t), H_{0,S}] \tag{A.13}$$

Furthermore, $H_{0,I} = H_{0,S}$. Equipped with this we can now calculate the expressions for operators in the interaction picture:

$$i\frac{da_I(t)}{dt} = [a_I(t), H_{0,S}] \tag{A.14}$$

$$= [a_I(t), \omega_c a^\dagger a] \tag{A.15}$$

$$= \omega_c a_I(t) \tag{A.16}$$

$$\Rightarrow \frac{da_I(t)}{dt} = -i\omega_c a_I(t) \rightarrow a_I(t) = ae^{-i\omega_c t} \tag{A.17}$$

Where we used that:

$$[a, H_{0,S}] = \omega_c[a, a^\dagger a] = \omega_c(\underbrace{[a, a^\dagger]}_{=1} a + a^\dagger \underbrace{[a, a]}_{=0}) = \omega_c a \tag{A.18}$$

Now we repeat the same thing for the atomic lowering operator σ:

$$i\frac{d\sigma_I(t)}{dt} = \underbrace{[\sigma_I(t), H_{0,S}]} \tag{A.19}$$

$$i\frac{d\sigma_I(t)}{dt} = \omega_a \sigma_I(t) \tag{A.20}$$

$$\Rightarrow \frac{d\sigma_I(t)}{dt} = -i\omega_a \sigma_I(t) \rightarrow \sigma_I(t) = \sigma e^{-i\omega_a t} \tag{A.21}$$

Now we can calculate the Hamiltonian for the interaction part and the drive part, in the interaction picture, by just plugging these operators in there, giving us:

$$H_{int,I} = \sigma a e^{-i(\omega_a - \omega_c)t} + \sigma a^\dagger e^{-i(\omega_a - \omega_c)t} + \sigma^\dagger a e^{-i(\omega_c - \omega_a)t} + \sigma^\dagger a^\dagger e^{i(\omega_c + \omega_a)t} \tag{A.22}$$

$$\xrightarrow{RWA} = \sigma a^\dagger e^{-i(\omega_a - \omega_c)t} + \sigma^\dagger a e^{-i(\omega_c - \omega_a)t} \tag{A.23}$$

$$H_{drive,I} = i(\epsilon e^{-i\omega_d t} + \epsilon^\dagger e^{i\omega_d t})(a_I(t) + a_I^\dagger(t)) \tag{A.24}$$

$$\xrightarrow{RWA} = i(a\epsilon^* e^{-i(\omega_c - \omega_d)t} + a^\dagger \epsilon e^{i(\omega_c - \omega_d)t}) \tag{A.25}$$

Thus giving us the total Hamiltonian in the interaction picture under the rotating wave approximation:

$$H_{JC,I} \overset{\text{RWA}}{=} \omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma + g(\sigma a^\dagger e^{-i(\omega_a - \omega_c)t} + \sigma^\dagger a e^{-i(\omega_c - \omega_a)t}) + i(a\epsilon^* e^{-i(\omega_c - \omega_d)t} + a^\dagger \epsilon e^{i(\omega_c - \omega_d)t}) \tag{A.26}$$

If we now transform back into the Schrödinger picture, we get:

$$H_{JC,S} \overset{\text{RWA}}{=} \omega_c a^\dagger a + \omega_a \sigma^\dagger \sigma + g(a^\dagger \sigma + a\sigma^\dagger) + i(a\epsilon^* e^{i\omega_d t} + a^\dagger \epsilon e^{-i\omega_d t}) \tag{A.27}$$

If we now change into the rotating frame of the driving field via

$$U(t) = \exp\{i\omega_d t(a^\dagger a + \sigma^\dagger \sigma)\} \tag{A.28}$$

$$H \to UHU^\dagger + i\dot{U}U^\dagger =: H_{RF} \tag{A.29}$$

we then get the final form of our Hamiltonian:

$$\boxed{H_{JC,RF} = \underbrace{(\omega_c - \omega_d)}_{=\Delta_c} a^\dagger a + \underbrace{(\omega_a - \omega_d)}_{=\Delta_a} \sigma^\dagger \sigma + g(a^\dagger \sigma + a\sigma^\dagger) + i(a\epsilon^* + a^\dagger \epsilon)} \tag{A.30}$$

Which gives rise to 4 scenarios which I'll be covering with my project:

scenario	**meaning**
$\Delta_a = \Delta_c = 0$	Atom is in coupled state (here: $\|1\rangle$ and we drive with a light field that's resonant to the cavity (here: we drive $\|1\rangle \leftrightarrow \|e\rangle$ which would be our $\|\pi\rangle$ polarized light
$\Delta_a = 0; \Delta_c \neq 0$	Atom is in coupled state but we drive with light that's non-resonant to cavity (here: $\|V\rangle$ polarized light)
$\Delta_a \neq 0; \Delta_c = 0$	Atom is in uncoupled state (here: $\|0\rangle$) and we drive with a light field that's resonant to the cavity
$\Delta_a, \Delta_c \neq 0$	Atom is in uncoupled state and we drive with light that's non-resonant to cavity

In order to improve readability we'll be splitting the full JC Hamiltonian into three parts:

$$H_0 = \Delta_c a^\dagger a + \Delta_a \sigma^\dagger \sigma \tag{A.31}$$

$$H_{int} = g(a^\dagger \sigma + a\sigma^\dagger) \tag{A.32}$$

$$H_{drive} = i(a\epsilon^* + a^\dagger \epsilon) \tag{A.33}$$

with following losses:

$$\gamma_1 = \gamma, \quad L_1 = \sigma, \tag{A.34}$$

$$\gamma_2 = \kappa, \quad L_2 = a. \tag{A.35}$$

A.3.3 Cavity mode a

For H_0:

$$[a, H_0] = \Delta_c[a, a^\dagger a] + \Delta_c a \underbrace{[a, \sigma^\dagger \sigma]}_{=0} \tag{A.36}$$

$$= \Delta_c(\underbrace{[a, a^\dagger]}_{=1} a + a^\dagger \underbrace{[a, a]}_{=0}) \tag{A.37}$$

$$= \Delta_c a \tag{A.38}$$

$$\boxed{\Rightarrow -i\langle[a, H_0]\rangle = -i\Delta_c \langle a\rangle} \tag{A.39}$$

For H_{int}:

$$[a, H_{int}] = g([a, a^\dagger \sigma] + [a, a\sigma^\dagger]) \tag{A.40}$$

$$= g(aa^\dagger \sigma - a^\dagger a\sigma + \cancel{aa\sigma^\dagger} - \cancel{aa\sigma^\dagger}) \tag{A.41}$$

$$\to \langle aa^\dagger \sigma\rangle = \langle n| aa^\dagger \sigma |n\rangle = (n+1)\langle\sigma\rangle \tag{A.42}$$

$$\to \langle a^\dagger a\sigma\rangle = \langle n| a^\dagger a\sigma |n\rangle = n\langle\sigma\rangle \tag{A.43}$$

$$\boxed{\Rightarrow -i\langle[a, H_{int}]\rangle = -ig\langle\sigma\rangle} \tag{A.44}$$

For H_{drive}:

$$\begin{aligned}
[a, H_{drive}] &= [a, i(a\epsilon^* + a^\dagger\epsilon)] &\text{(A.45)}\\
&= i([a, a\epsilon^*] + [a, a^\dagger\epsilon]) &\text{(A.46)}\\
&= i(0 + \epsilon) &\text{(A.47)}\\
&= i\epsilon &\text{(A.48)}
\end{aligned}$$

$$\boxed{\Rightarrow -i\langle[a, H_{drive}]\rangle = i\langle\epsilon\rangle \overset{\epsilon\in\mathbb{C}}{=} i\epsilon} \tag{A.49}$$

If we now include the fact that:

$$i\epsilon \mathrel{\widehat{=}} \sqrt{\kappa_{oc}}a^{in}(t) \tag{A.50}$$

we are then left with:

$$\boxed{\Rightarrow -i\langle[a, H_{drive}]\rangle = -\sqrt{\kappa_{oc}}a^{in}(t)} \tag{A.51}$$

For rest of the Lindbladian:

$$\begin{aligned}
D(a) &= a^\dagger a a - \frac{1}{2}\{a^\dagger a, a\} &\text{(A.52)}\\
&= a^\dagger a a - \frac{1}{2}(a^\dagger a a - a a^\dagger a) &\text{(A.53)}\\
&= \frac{1}{2}a^\dagger a a - \frac{1}{2}a a^\dagger a &\text{(A.54)}\\
&= \frac{1}{2}(a^\dagger a a - a a^\dagger a) &\text{(A.55)}\\
&= \frac{1}{2}([a^\dagger, a]a) &\text{(A.56)}\\
&= \frac{1}{2}(-a) &\text{(A.57)}\\
&= -\frac{1}{2}a &\text{(A.58)}
\end{aligned}$$

$$\boxed{\Rightarrow \kappa\langle D(a)\rangle = -\frac{\kappa}{2}\langle a\rangle} \tag{A.59}$$

So for $\langle a\rangle(t)$ we get:

$$\boxed{\begin{aligned}
\frac{d\langle a\rangle}{dt} &= -i\Delta_c\langle a\rangle - ig\langle\sigma\rangle - \sqrt{\kappa_{oc}}\,a_{\text{in}}(t) - \frac{\kappa}{2}\langle a\rangle\\
&= -(i\Delta_c + \tfrac{\kappa}{2})\langle a\rangle - ig\langle\sigma\rangle - \sqrt{\kappa_{oc}}\,a_{\text{in}}(t)
\end{aligned}} \tag{A.60}$$

A.3.4 Atomic lowering operator σ

For H_0:

$$\begin{aligned}
[\sigma, H_0] &= \Delta_a \underbrace{[\sigma, \sigma^\dagger\sigma]}_{=\underbrace{[\sigma,\sigma^\dagger]}_{=\mathbb{1}}\sigma + \sigma^\dagger\underbrace{[\sigma,\sigma]}_{=0}} &\text{(A.61)}\\
&= \Delta_a\mathbb{1}\sigma &\text{(A.62)}
\end{aligned}$$

$$\boxed{\Rightarrow -i\langle[\sigma, H_0]\rangle = -i\Delta_a\langle\sigma\rangle} \tag{A.63}$$

For H_{int}:

$$
\begin{aligned}
[\sigma, H_{int}] &= g([\sigma, a^\dagger\sigma] + [\sigma, a\sigma^\dagger]) && \text{(A.64)}\\
&= g(\cancel{a^\dagger\sigma\sigma} - \cancel{a^\dagger\sigma\sigma} + a\sigma\sigma^\dagger - a\sigma^\dagger\sigma) && \text{(A.65)}\\
&= g(a\sigma\sigma^\dagger - a\sigma^\dagger\sigma) && \text{(A.66)}
\end{aligned}
$$

Now we can consider that:

$$
\boxed{\begin{aligned}
\sigma^\dagger\sigma &= |e\rangle\langle 1|\langle e| = |e\rangle\langle e| \\
\sigma\sigma^\dagger &= |1\rangle\langle e|\langle 1| = |1\rangle\langle 1|
\end{aligned}}
\quad \begin{aligned}&\text{(A.67)}\\&\text{(A.68)}\end{aligned}
$$

$$
\begin{aligned}
[\sigma, H_{int}] &= g(a\sigma\sigma^\dagger - a\sigma^\dagger\sigma) && \text{(A.69)}\\
&= ga(|1\rangle\langle 1| - |e\rangle\langle e|) && \text{(A.70)}\\
&= ga\sigma_z && \text{(A.71)}
\end{aligned}
$$

$$
\boxed{\Rightarrow -i\langle[\sigma, H_{int}]\rangle = -ig\langle a\sigma_z\rangle} \tag{A.72}
$$

No interaction with H_{drive}.
For rest of Lindbladian:

$$
\begin{aligned}
D(\sigma) &= \underbrace{\sigma^\dagger \underbrace{\sigma\sigma}_{=0}}_{=0} - \frac{1}{2}\{\sigma^\dagger\sigma, \sigma\} && \text{(A.73)}\\
&= -\frac{1}{2}\underbrace{\sigma^\dagger\sigma\sigma}_{=0} - \frac{1}{2}\sigma\sigma^\dagger\sigma && \text{(A.74)}\\
&= -\frac{1}{2}\sigma && \text{(A.75)}
\end{aligned}
$$

$$
\boxed{\gamma\langle D(\sigma)\rangle = -\frac{\gamma}{2}\langle\sigma\rangle} \tag{A.76}
$$

So for $\langle\sigma\rangle$(t) we get:

$$
\boxed{\begin{aligned}
\frac{d\langle\sigma\rangle}{dt} &= -i\Delta_a\langle\sigma\rangle - ig\langle a\sigma_z\rangle - \frac{\gamma}{2}\langle\sigma\rangle \\
&= -(i\Delta_a + \frac{\gamma}{2})\langle\sigma\rangle - ig\langle a\sigma_z\rangle
\end{aligned}} \tag{A.77}
$$

A.3.5 Pauli-Z operator σ_z

For H_0:

$$
\begin{aligned}
[\sigma_z, H_0] &= \Delta_a[\sigma_z, \sigma^\dagger\sigma] && \text{(A.78)}\\
&= \Delta_a[\sigma_z, |e\rangle\langle e|] && \text{(A.79)}\\
&= \Delta_a(\sigma_z|e\rangle\langle e| - |e\rangle\langle e|\sigma_z) && \text{(A.80)}\\
&= \Delta_a(-|e\rangle\langle e| + |e\rangle\langle e|) && \text{(A.81)}\\
&= 0 && \text{(A.82)}
\end{aligned}
$$

For H_{int}:

$$
\begin{aligned}
[\sigma_z, H_{int}] &= g([\sigma_z, a^\dagger\sigma] + [\sigma_z, a\sigma^\dagger]) && \text{(A.83)}\\
&= g(a^\dagger\sigma_z\sigma - a^\dagger\sigma\sigma_z + a\sigma_z\sigma^\dagger - a\sigma^\dagger\sigma_z) && \text{(A.84)}\\
&= g(a^\dagger\sigma + a^\dagger\sigma - a\sigma^\dagger - a\sigma^\dagger) && \text{(A.85)}\\
&= 2g(a^\dagger\sigma - a\sigma^\dagger) && \text{(A.86)}
\end{aligned}
$$

$$\boxed{\Rightarrow -i\langle[\sigma_z, H_{int}]\rangle = -2ig(\langle a^\dagger\sigma\rangle - \langle a\sigma^\dagger\rangle)} \tag{A.87}$$

No interaction with H_{drive}. For rest of Lindbladian:

$$D(\sigma_z) = \sigma^\dagger\sigma_z\sigma - \frac{1}{2}\{\sigma^\dagger\sigma, \sigma\} \tag{A.88}$$

$$= |e\rangle\langle e| - \frac{1}{2}\underbrace{\sigma^\dagger\sigma\sigma_z}_{=-|e\rangle\langle e|} - \frac{1}{2}\underbrace{\sigma_z\sigma^\dagger\sigma}_{=-|e\rangle\langle e|} \tag{A.89}$$

$$= |e\rangle\langle e| + |e\rangle\langle e| \tag{A.90}$$

$$= 2|e\rangle\langle e| \tag{A.91}$$

$$= \mathbb{1} - \sigma_z \tag{A.92}$$

$$\boxed{\Rightarrow \gamma\langle D(\sigma_z)\rangle = \gamma(1 - \langle\sigma_z\rangle)} \tag{A.93}$$

So for $\langle\sigma_z\rangle(t)$ we get:

$$\boxed{\frac{d\langle\sigma_z\rangle(t)}{dt} = -2ig(\langle a^\dagger\sigma\rangle - \langle a\sigma^\dagger\rangle) + \gamma(1 - \langle\sigma_z\rangle)} \tag{A.94}$$

The final Maxwell Bloch Equations then take the form of:

$$\frac{d\langle a\rangle}{dt} = -(i\Delta_c + \tfrac{\kappa}{2})\langle a\rangle - ig\langle\sigma\rangle - \sqrt{\kappa_{oc}}\, a_{\text{in}}(t) \tag{A.95}$$

$$\frac{d\langle\sigma\rangle}{dt} = -(i\Delta_a + \tfrac{\gamma}{2})\langle\sigma\rangle - ig\langle a\sigma_z\rangle \tag{A.96}$$

$$\frac{d\langle\sigma_z\rangle}{dt} = -2ig\big(\langle a^\dagger\sigma\rangle - \langle a\sigma^\dagger\rangle\big) + \gamma\big(1 - \langle\sigma_z\rangle\big) \tag{A.97}$$

Bibliography

[1] L.-M. Duan and H. J. Kimble. "Scalable Photonic Quantum Computation through Cavity-Assisted Interactions". In: *Phys. Rev. Lett.* 92 (12 Mar. 2004), p. 127902. DOI: 10.1103/PhysRevLett.92.127902. URL: https://link.aps.org/doi/10.1103/PhysRevLett.92.127902.

[2] Andreas Reiserer et al. "A quantum gate between a flying optical photon and a single trapped atom". In: *Nature* 508 (7495 Apr. 2014), pp. 237–240. ISSN: 14764687. DOI: 10.1038/nature13177. URL: https://www.nature.com/articles/nature13177.

[3] Andreas Reiserer and Gerhard Rempe. "Cavity-based quantum networks with single atoms and optical photons". In: *Rev. Mod. Phys.* 87 (4 Dec. 2015), pp. 1379–1418. DOI: 10.1103/RevModPhys.87.1379. URL: https://link.aps.org/doi/10.1103/RevModPhys.87.1379.

[4] E.T. Jaynes and F.W. Cummings. "Comparison of quantum and semiclassical radiation theories with application to the beam maser". In: *Proceedings of the IEEE* 51.1 (1963), pp. 89–109. DOI: 10.1109/PROC.1963.1664.

[5] E.T. Jaynes and F.W. Cummings. "Comparison of quantum and semiclassical radiation theories with application to the beam maser". In: *Proceedings of the IEEE* 51.1 (1963), pp. 89–109. DOI: 10.1109/PROC.1963.1664.

[6] Manuel K. L. Brekenfeld. "Ein photonischer Quantenspeicher mit einem Atom in gekreuzten optischen Faserresonatoren". de. Technische Universität München, 2023, p. 184. URL: https://mediatum.ub.tum.de/1695291.

[7] Manuel Uphoff et al. "Frequency splitting of polarization eigenmodes in microscopic FabryPerot cavities". In: *New Journal of Physics* 17.1 (Jan. 2015), p. 013053. DOI: 10.1088/1367-2630/17/1/013053. URL: https://doi.org/10.1088/1367-2630/17/1/013053.

[8] C. W. Gardiner and M. J. Collett. "Input and output in damped quantum systems: Quantum stochastic differential equations and the master equation". In: *Phys. Rev. A* 31 (6 June 1985), pp. 3761–3774. DOI: 10.1103/PhysRevA.31.3761. URL: https://link.aps.org/doi/10.1103/PhysRevA.31.3761.

[9] C. Y. Hu et al. "Giant optical Faraday rotation induced by a single-electron spin in a quantum dot: Applications to entangling remote spins via a single photon". In: *Phys. Rev. B* 78 (8 Aug. 2008), p. 085307. DOI: 10.1103/PhysRevB.78.085307. URL: https://link.aps.org/doi/10.1103/PhysRevB.78.085307.

[10] T. G. Tiecke et al. "Nanophotonic quantum phase switch with a single atom". In: *Nature* 508.7495 (Apr. 2014), pp. 241–244. ISSN: 1476-4687. DOI: 10.1038/nature13188. URL: https://doi.org/10.1038/nature13188.

[11] Dominik Niemietz. "Nondestructive detection of photonic qubits with single atoms in crossed fiber cavities". en. Technische Universität München, 2021, p. 113. URL: https://mediatum.ub.tum.de/1601857.

[12] D. F. Walls and G. J. Milburn. *Quantum Optics*. 2nd. Berlin, Heidelberg: Springer, 2008. ISBN: 978-3-540-28574-8. DOI: 10.1007/978-3-540-28574-8.

Selbständigkeitserklärung

Ich versichere hiermit, dass ich die von mir eingereichte Abschlussarbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

gezeichnet,
Leart Zuka, München den TBD.